

Math

Calculus

1. Find derivative with respect to x of the function $f(x) = \ln \sqrt{\frac{1-\sin x}{1+\sin x}}$ at $x \neq \pm\pi/2 + \pi n, n \in \mathbf{Z}$

- ☐ $-1/\sin x$
- ☒ $-1/\cos x$
- ☐ $1/\cos x$
- ☐ $\sin x$
- ☐ $-\cos x$

2. Find the definite integral $\int_0^1 \frac{x^2 + 3x}{(x+1)(x^2+1)} dx$

- ☐ π
- ☐ $\pi/2$
- ☒ $\pi/4$
- ☐ π^2
- ☐ $\pi^2/2$
- ☐ $\pi^2/4$

3. Find the limit $\lim_{x \rightarrow 0} \frac{1 - \sqrt{\cos x}}{x - x \cos \sqrt{x}}$

• $\frac{1}{2}$

◦ $-\frac{1}{4}$

◦ 2

◦ -1

◦ -1/2

◦ 1/4

4. Calculate partial derivatives of the function $f(x, y) = \sqrt[3]{\sin^4 x + \cos^4 y}$ at $x = \pi/4$, $y = \pi/4$

◦ $\partial_x = \frac{2^{1/3}}{3}$, $\partial_y = \frac{2^{2/3}}{3}$

◦ $\partial_x = -\frac{2^{2/3}}{3}$, $\partial_y = -\frac{2^{2/3}}{3}$

◦ $\partial_x = -\frac{2^{1/3}}{3}$, $\partial_y = \frac{2^{1/3}}{3}$

◦ $\partial_x = -\frac{2^{2/3}}{3}$, $\partial_y = -\frac{2^{1/3}}{3}$

• $\partial_x = \frac{2^{2/3}}{3}$, $\partial_y = -\frac{2^{2/3}}{3}$

5. Find the first five terms (up to x^3) of the Taylor series expansion around $x = 0$ of the function $f(x) = (1+x)^{1/x}$

• $e - \frac{ex}{2} + \frac{11ex^2}{24} - \frac{7ex^3}{16} + O(x^4)$

◦ $e - \frac{ex}{2} + \frac{11ex^2}{24} + \frac{7ex^3}{16} + O(x^4)$

◦ $e + \frac{ex}{2} + \frac{11ex^2}{24} + \frac{7ex^3}{16} + O(x^4)$

◦ $e/2 + \frac{ex}{4} + \frac{11ex^2}{48} + \ln \frac{7ex^3}{32} + O(x^4)$

◦ $2e - ex + \frac{11ex^2}{12} - \frac{7ex^3}{8} + O(x^4)$

Differential equations

10. Determine type of the differential equation $u_t = xu_{xx} + u_{yy}$

- first order differential equation
- linear differential equation with constant coefficients
- linear nonhomogeneous differential equation
- nonlinear nonhomogeneous differential equation
- nonlinear homogeneous differential equation
- none of the above

11. Write general solution of the differential equation $x^2y'' + a_2xy' + a_3y = 0$

- $Ax^2 + Bx + C$
- $Ax^{\lambda_1} + Bx^{\lambda_2} + Cx^{\lambda_3}$
- $Ax^{\lambda_1} + Bx^{\lambda_2}$
- $Ae^{\lambda_1x} + Be^{\lambda_2x}$
- $Ae^{\lambda x} + Bxe^{\lambda x}$
- explicit algebraic/trigonometric form does not exist

12. Solve the differential equation $xy' + 2y = 2\cos x$

- $(x \cos x + 2 \sin x + C)/x^2$
- $(x \cos x - 2 \sin x + C)/x^2$
- $(2x \cos x + 2 \sin x + C)/x^2$
- $(2x \cos x - 2 \sin x + C)/x^2$
- $(2x \sin x + 2 \cos x + C)/x^2$
- $(2x \sin x - 2 \cos x + C)/x^2$

13. For the equation $xy' + 2y = 2 \cos x$ determine $y(\pi)$ if $y(0) = 1$

- ☒ $-4/\pi^2$
- ☐ $-2/\pi^2$
- ☐ $-1/\pi^2$
- ☐ $1/\pi^2$
- ☐ $2/\pi^2$
- ☐ $4/\pi^2$

Linear algebra

14. Let $\mathbf{A} = \begin{pmatrix} 6 & -5 \\ 7 & -6 \end{pmatrix}$. Find $\mathbf{A}^{-1}$

☐ $\begin{pmatrix} 6 & 5 \\ -7 & -6 \end{pmatrix}$

☐ $\begin{pmatrix} -6 & -5 \\ 7 & 6 \end{pmatrix}$

☐ $\begin{pmatrix} -6 & 5 \\ -7 & 6 \end{pmatrix}$

☒ $\begin{pmatrix} 6 & -5 \\ 7 & -6 \end{pmatrix}$

☐ None of the above

15. Let $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$ be square invertible matrices of the same size. Simplify the expression

$$(\mathbf{A}^6 \mathbf{B}^3 \mathbf{C}^{-2})^{-1} \mathbf{A}^2 (\mathbf{C}^2 \mathbf{B}^{-3} \mathbf{A}^{-4})^{-1}$$

☒ $\mathbf{I}$ (identity matrix)

☐ $\mathbf{A}$

☐ $\mathbf{B}$

☐ $\mathbf{C}$

☐ None of the above

16. Let $\mathbf{A}$ be a 5×5 real matrix such that

$$\det(2 \mathbf{A}^2 (\mathbf{A}^T)^2 \mathbf{A}^{-1}) = 4,$$

where $\mathbf{A}^T$ is the transpose matrix. Find $\det \mathbf{A}$

☐ 0

☐ 1

☒ $1/2$

☐ 2

☐ 4

☐ None of the above

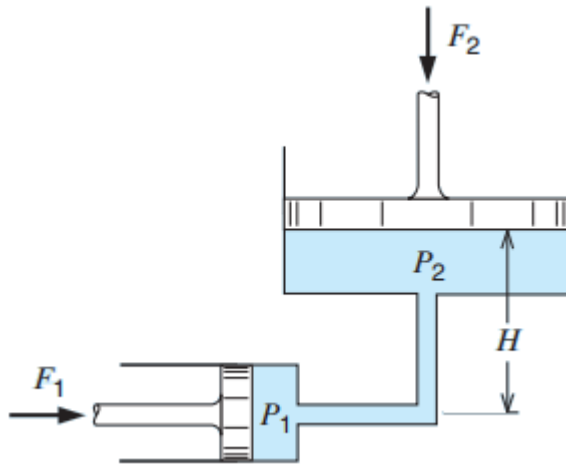
Probability

19. For a random permutation of 4 objects, what is the probability that no object remains at its place? Express the result as an irreducible fraction a/b with integer a, b .

20. Find the variance of a random variable X such that $P(X = 0) = \frac{1}{3}$, $P(X = 2) = \frac{2}{3}$. Express the result as an irreducible fraction a/b with integer a, b .

Physics

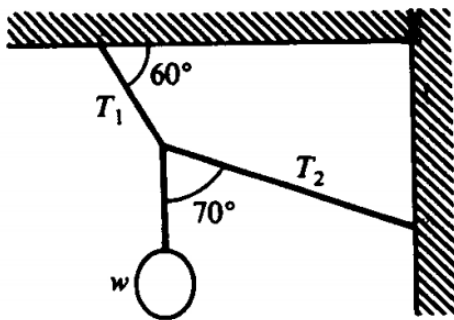
Problem 1



Assume a system in the picture: a cylinder with a cross-sectional area of 0.01 m^2 is connected to a bigger cylinder with a cross-sectional area of 0.05 m^2 . Both cylinders and line are filled with a liquid of density 900 kg/m^3 . Larger cylinder is $H = 6 \text{ m}$ higher up in elevation. Outside atmospheric pressure is 100 kPa and a net force on the smallest cylinder is $F_1 = 25 \text{ kN}$. What is the balancing force F_2 on the second larger cylinder?

- a) 5 kN b) 122.4 kN c) 125 kN d) 132.2 kN e) 127.6 kN

Problem 2



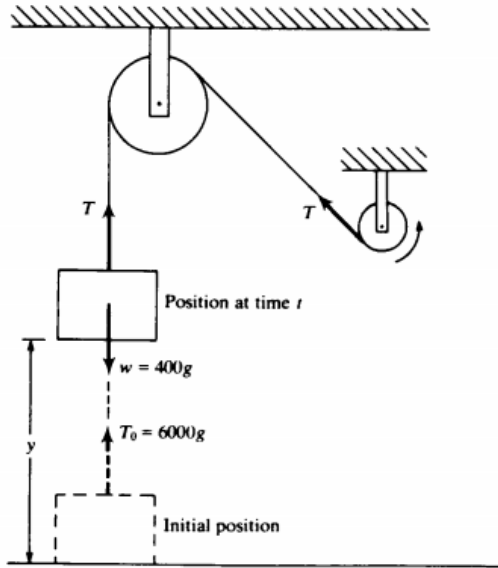
Consider the system in the picture. Find T_1 and T_2 , if $w = 40 \text{ N}$ and system is in the equilibrium.

- a) $T_1 = 37.6 \text{ N}$; $T_2 = 13.7 \text{ N}$ b) $T_1 = 22.5 \text{ N}$; $T_2 = 68.9 \text{ N}$ c) $T_1 = 21.3 \text{ N}$; $T_2 = 53.9 \text{ N}$
 d) $T_1 = 38.2 \text{ N}$; $T_2 = 20.3 \text{ N}$ e) $T_1 = 58.4 \text{ N}$; $T_2 = 31.1 \text{ N}$

Problem 3

A body of mass 400 kg is suspended at the lower end of a light chain and is being pulled up vertically (see the figure below). Initially body is at rest and the pull on the chain is 6000 g . The pull gets smaller uniformly at the rate of 360 g per each meter through which the body is raised. What is the velocity of the body when it has been raised 10 m ? ($g = 9.8 \text{ m/s}^2$)

- a) 17.6 m/s b) 54.2 m/s c) 22.4 m/s d) 52.4 m/s e) 43.2 m/s

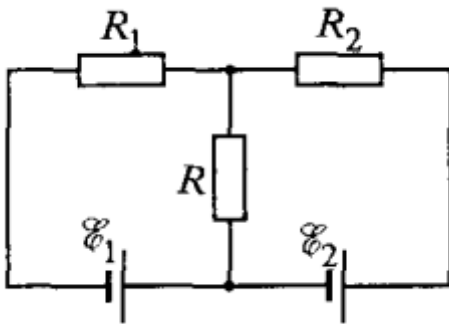


Problem 4

Find an absolute change of internal energy ΔU of nitrogen mass while its adiabatic expanding from volume $V_1 = 0.01 \text{ m}^3$ taken at normal pressure $P_1 = 100 \text{ kPa}$ to volume $V_2 = 0.32 \text{ m}^3$.

- a) 1875 J b) 0 J c) 663 J d) 2860 J e) 326 J

Problem 5



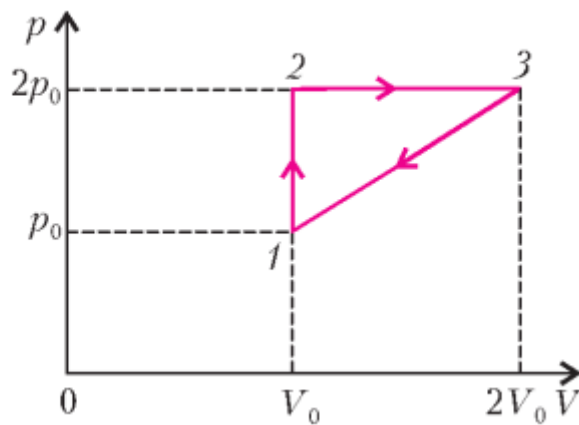
Consider electrical scheme in the picture with resistors $R_1 = 10 \Omega$ and $R_2 = 15 \Omega$. Find resistance R at which the power dissipated on it is maximum. $\mathcal{E}_1 = 5\text{V}$, $\mathcal{E}_2 = 3\text{V}$.

- a) 19.4Ω b) 25Ω c) 6Ω d) 3.2Ω e) 1.9Ω

Problem 6

The tank is kept filled with water to a depth of 8.0 m. Find the speed v with which the jet of water emerges from the small pipe just at the bottom of the tank.

- a) 8.9 m/s b) 6.3 m/s c) 17.7 m/s d) 11 m/s e) 12.5 m/s

Problem 7

Find the efficiency of the cycle in the picture.

- a) 5.2% b) 11.1% c) 6.3% d) 7.7% e) 25%

Problem 8

How far should an object be from a concave mirror of radius 36 cm to form a real image one-ninth its size?

- a) 720 cm b) 360 cm c) 240 cm d) 180 cm e) 90 cm

Problem 9

Consider LC circuit. At some moment, dielectric plate with factor $\varepsilon = 4$ with is removed from the capacitor very quickly. Find the relative change of the oscillation frequency ω/ω_0 .

- a) 8 b) 4 c) 2 d) 1 e) 0.25

Problem 10

Given that ionization energy of the hydrogen atom is 13.6 eV, calculate the wavelength of the first line of the Lyman series.

- a) 121.6 nm b) 72.9 nm c) 182.4 nm d) 91.1 nm e) 364.7 nm

Chemistry

1) The electrical conductivities of 0.10 M solutions of NaCl, HCN ($K_a = 4.9 \times 10^{-10}$) and HNO_2 ($K_a = 4.6 \times 10^{-4}$) are measured. The order by conductivity from highest to lowest is

A – $\text{NaCl} > \text{HNO}_2 > \text{HCN}$

B – $\text{HCN} > \text{HNO}_2 > \text{NaCl}$

C – $\text{NaCl} > \text{HCN} > \text{HNO}_2$

D – $\text{HNO}_2 > \text{HCN} > \text{NaCl}$

2) Consider the redox reaction $\text{Zn}^0(\text{s}) + 2\text{H}^+(\text{aq}) \rightarrow \text{Zn}^{2+}(\text{aq}) + \text{H}_2^0(\text{g})$. What species are reduced in the process?

A – Zn; B – H^+ ; C – H_2 ; D – none

3) Range the hydrocarbons A, B, C, D according to their retention times on chromatographic column with the non-polar stationary phase (squalane):

Compound		Molecular formula	Boiling point, °C
A	n-pentane	C_5H_{12}	36°C
B	2-methylbutane	C_5H_{12}	28°C
C	n-hexane	C_6H_{14}	68°C
D	thiophene	$\text{C}_4\text{H}_4\text{S}$	84°C

A – $\text{C} \rightarrow \text{D} \rightarrow \text{A} \rightarrow \text{B}$;

B – $\text{A} \rightarrow \text{D} \rightarrow \text{C} \rightarrow \text{B}$;

C – $\text{B} \rightarrow \text{A} \rightarrow \text{D} \rightarrow \text{C}$;

D – $\text{B} \rightarrow \text{A} \rightarrow \text{C} \rightarrow \text{D}$

4) Bromine has an oxidation number of +1 in

A – NaBrO; B – NaBrO₂; C – NaBrO₃; D – NaBrO₄

5) How many grams of CO_2 (MW = 44 g/mol) is required for the reaction with $\text{Ba}(\text{OH})_2$ to yield 0.4 g of BaCO_3 (MW = 197 g/mol)?

A – 0.4 g; B – 0.22 g; C – 0.07 g; D – 0.09 g

6) Which of the following are isomers?

A – Fe^{2+} and Fe^{3+}

B – O_2 and O_3

C – iso-pentane and n-pentane

D – hydrogen and deuterium

7) Consider the equilibrium $\text{CO}(\text{gas}) + \text{H}_2\text{O}(\text{gas}) \leftrightarrow \text{CO}_2(\text{gas}) + \text{H}_2(\text{gas})$. What will favor **the forward reaction**?

A – a pressure increase; B – a pressure decrease; C – the addition of CO_2 ; D – the addition of CO

8) To remove elemental sulfur from the solution activated copper is added to a flask. Which factor would account for the faster desulfurization reaction rate?

A – air bubbling;

B – iron oxide addition;

C – preliminary shredding of the copper;

D – cooling of the flask.

9) Choose the emulsion from the list below:

A – Milk; B – Smoke; C – Ice; D – Carbonated water

10) Calculate C/H ratio (mass parts ratio of carbon to hydrogen) for anthracene ($\text{C}_{14}\text{H}_{10}$)?

A - 14; B – 1.4; C - 12; D – 16.8

Geology

- 1 In what climate zone would you expect accumulation of carbonate deposits?
 - A Tropical zone
 - B Subpolar zone
 - C Temperate zone
- 2 What is the correct definition of a depositional sequence ?
 - A A layer deposited during the same geological time
 - B A layer composed of similar lithology
 - C A layer of deposits, accumulated in similar sedimentation environment.
- 3 What rock have the finer average grain size?
 - A Siltstone
 - B Sandstone
 - C Mudstone
- 4 What lithology correspond to spontaneous potential log base line
 - A Sandstone
 - B Dolomite
 - C Claystone
- 5 What well log is used for water saturation evaluation?
 - A Spontaneous potential log
 - B Resistivity log
 - C Gamma log
- 6 Combination of what logs can help to identify presence of gas in a reservoir?
 - A Resistivity and Neutron log
 - B Spontaneous potential and gamma log
 - C Density and neutron log
- 7 What types of contacts dominate in clay deposits?
 - A Chemical
 - B Colloidal
 - C Cementation
- 8 What laboratory test is based on Darcy's law?
 - A Fluid permeability measurement

B Electrical current measurement

C Porosity measurement

9 In what rock would you predict higher value of water-oil transition zone based on the following mercury porosimetry data?

A Median pore throat size = 1 micrometer

B Median pore throat size = 10 micrometers

C Median pore throat size = 100 micrometers

10 Presume that ramp depositional model is applied for our reservoir. Please select the most probable microfacies trend from proximal to distal direction.

A Wackestone, Packstone, Grainstone, Packstone, Wackestone.

B Wackestone, Mudstone, Grainstone, Mudstone, Grainstone.

C Boundstone, Mudstone, Grainstone, Boundstone.